SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA0401**

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01 **Total Marks:100**

Annexure A

CASE STUDY

Code of Conduct:

*I __**A.Sai Rohit (192472144)**__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Sai Rohit

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Levels**	**Marks**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 8: Food Delivery Service Analysis
Scenario:
A food delivery company has collected order data, including delivery time, order value, customer rating, distance travelled, and delivery partner experience. The company wants to reduce delivery delays and improve customer satisfaction.
Questions:
A. Explain how descriptive statistics such as mean, median, standard deviation, and quartiles can be used to analyze delivery time and customer ratings.
B. Discuss how unusually long delivery times can affect customer satisfaction analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether experienced delivery partners complete deliveries faster than new delivery partners? Formulate the hypotheses and explain the analysis process.

TITLE

Food Delivery Service Analysis Using Descriptive Statistics, Outlier Detection and Hypothesis Testing

INTRODUCTION

Food delivery services generate a large amount of data from daily orders. This data contains useful information such as delivery time, order value, customer rating, distance travelled and delivery partner experience. Analyzing this information helps a company understand its delivery performance and customer satisfaction.

Delivery time is one of the most important factors in food delivery because customers generally expect their orders to arrive within the promised time. Long delivery times can result in poor customer ratings and dissatisfaction. Therefore, statistical analysis can be used to identify normal delivery patterns and unusual delays.

In this case study, descriptive statistical techniques such as mean, median, standard deviation and quartiles are used to summarize the delivery data. The Interquartile Range (IQR) and Z-score methods are discussed for identifying outliers. Distribution analysis and graphs are used to understand how delivery times are spread across different orders.

Finally, hypothesis testing is performed to determine whether experienced delivery partners complete deliveries faster than new delivery partners. An independent two-sample t-test is used to compare the delivery times of the two groups.

The results can help the company identify delivery problems, improve delivery partner performance and increase customer satisfaction.

PROBLEM STATEMENT

A food delivery company has collected order-related data containing delivery time, order value, customer rating, distance travelled and delivery partner experience. The company is facing problems related to delivery delays and variations in customer satisfaction.

The company needs to analyze the collected data to understand the normal delivery time, identify unusually long deliveries and determine whether delivery partner experience has an impact on delivery performance.

The problem is to use statistical techniques to:

1. Summarize the delivery data using descriptive statistics.
2. Analyze the distribution of delivery times and customer ratings.
3. Detect unusually long delivery times.
4. Understand how extreme delivery times can affect customer satisfaction analysis.
5. Compare experienced and new delivery partners.
6. Determine statistically whether experienced delivery partners complete deliveries faster.
7. Provide useful recommendations to reduce delivery delays.

OBJECTIVES

The objectives of this case study are:

1. To calculate the mean delivery time.
2. To calculate the median delivery time.
3. To calculate the standard deviation of delivery time.
4. To determine the quartiles and interquartile range.
5. To identify unusually long delivery times using the IQR method.
6. To analyze customer rating statistics.
7. To understand the distribution of delivery times using graphs.
8. To study the effect of long delivery times on customer satisfaction.
9. To compare experienced and new delivery partners.
10. To perform hypothesis testing using an independent two-sample t-test.
11. To determine whether delivery partner experience significantly affects delivery speed.
12. To provide recommendations for improving food delivery services.

DATASET DESCRIPTION

A sample dataset containing 20 food delivery orders is considered for the analysis. Each observation represents one completed food delivery order.

The dataset contains the following attributes:

Attribute	Description
Order ID	Unique identification number of the order
Delivery Time	Time taken to deliver the order in minutes
Order Value	Total value of the food order in rupees
Customer Rating	Rating given by the customer from 1 to 5
Distance	Distance travelled by the delivery partner in kilometres
Experience	Delivery partner's experience in years

Variables

Numerical variables:

- Delivery Time
- Order Value
- Customer Rating
- Distance
- Experience

Important variables for this analysis:

- Delivery Time
- Customer Rating
- Experience

TOOLS AND TECHNOLOGIES

The following tools and technologies are used:

- **Python** – Programming language used for statistical analysis.
- **Pandas** – Used for creating and manipulating the dataset.
- **NumPy** – Used for numerical calculations.
- **SciPy** – Used for statistical hypothesis testing.
- **Matplotlib** – Used for generating graphs and visualizations.
- **Jupyter Notebook / Google Colab / VS Code** – Used to execute the Python program.

STATISTICAL METHODS USED

The following statistical techniques are used in this case study:

Descriptive Statistics

1. Mean
2. Median
3. Standard deviation
4. Minimum
5. Maximum
6. Quartiles

Outlier Detection

1. Interquartile Range method
2. Z-score method
3. Box plot

Distribution Analysis

1. Histogram
2. Box plot
3. Scatter plot

Statistical Inference

1. Hypothesis formulation
2. Independent two-sample t-test
3. p-value based decision

TEST USED

An independent two-sample t-test is used because we are comparing the delivery times of two independent groups:

- Experienced delivery partners
- New delivery partners

The test is one-tailed because we specifically want to know whether experienced partners are faster.

Dataset:

Order ID	Delivery Time (min)	Order Value (₹)	Rating	Distance (km)	Experience (years)
1	28	450	4.5	4.2	3
2	32	520	4.2	5.1	2
3	25	350	4.8	3.2	4
4	30	600	4.4	4.8	3
5	35	420	4.0	6.0	1
6	27	550	4.6	3.8	5
7	40	300	3.8	7.2	1
8	29	480	4.3	4.5	2
9	31	700	4.1	5.0	3
10	26	390	4.7	3.5	4
11	33	500	4.0	5.8	1
12	24	450	4.9	2.9	5
13	36	620	3.9	6.5	2
14	28	350	4.5	4.0	3
15	45	800	3.5	8.0	1
16	30	470	4.4	4.6	4
17	27	530	4.6	3.7	5
18	34	410	4.1	5.5	2
19	29	580	4.5	4.3	3
20	60	450	2.8	10.0	1

Calculations:

(1) Mean $= \frac{\Sigma x}{n}$

$$\frac{\Sigma x}{n} = \frac{28+32+35+30+35+27+40+29+31+26+33+24+36+28+45+30+27+34+29+60}{20}$$

$$= \frac{649}{20}$$

$$\boxed{= 32.45 \text{ minutes}}$$

Median $= \frac{10^{th} + 11^{th}}{2}$

$$= \frac{30+30}{2}$$

$$\boxed{\text{median} = 30 \text{ minutes}}$$

Range = max − min

$$= 60 - 24$$

$$\boxed{= 36 \text{ minutes}}$$

Standard deviation :

$$S.D = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n-1}}$$

$$\boxed{S.D = 8 \text{ minutes}}$$

$\bar{x} = 32.45$

$\Sigma (x-\bar{x})^2 = 1210.95$

$$S = \sqrt{\frac{1210.95}{19}}$$

$$\boxed{= 7.98}$$

Quartiles

$Q_1 = 27.75$ $\longrightarrow$ $Q_1 = n+1/4 = 21/4 \Rightarrow 5.25$

$Q_1 = 27 + 0.25 (28-27)$

$$\boxed{Q_1 = 21.75}$$

$Q_2 = 30$

$Q_3 = 34.25$

IQR $= Q_3 - Q_1$

$= 34.25 - 27.75$

$$\boxed{IQR = 6.5}$$

lower bound $= Q_1 - 1.5(IQR)$

$= 27.75 - 1.5(6.5)$

$= 27.75 - 9.75$

$\boxed{\text{lower bound} = 18}$

upper bound $= Q_3 + 1.5(IQR)$

$= 34.25 + 1.5(6.5)$

$= 34.25 + 9.75$

$\boxed{\text{upper bound} = 44}$

Outliner

$18 \leq x \leq 44$

$x < 18$ & $x > 44$

45 min = outliner

60 min = outliner

mean customer rating:

$\Sigma x = 84.6$

$n = 20$

mean $= \frac{\Sigma x}{n} = \frac{84.6}{20}$

$\boxed{\text{mean} = 4.23}$

Hypothesis testing:

exp $\geq$ 3 yrs

$n_1 = 11$

$\bar{x}_1 = \frac{305}{11}$

$\boxed{\bar{x}_1 = 27.73 \text{ minute}}$

new Partners

exP < 3yrs

$n_2 = 9$

$\bar{x}_2 = \frac{344}{9}$

$\boxed{\bar{x}_2 = 38.22 \text{ min}}$

$27.73 < 38.22$.

Null Hypothesis: $H_0 : \mu_E \geq \mu_N$

alternative hypothesis : $H_1 : \mu_E < \mu_N$

$\boxed{\alpha = 0.05}$

$$t = \frac{\bar{x}_E - \bar{x}_N}{\sqrt{\frac{S_E^2}{n_E} + \frac{S_N^2}{n_N}}}$$

$\bar{x}_E = 27.73$, $\bar{x}_N = 38.22$

$$t = \frac{27.73 - 38.22}{\sqrt{\frac{(2.20)^2}{11} + \frac{(9.40)^2}{9}}}$$

$\boxed{t = -3.28}$ $\boxed{P = 0.00501}$ $\boxed{\alpha = 0.05}$

$0.00501 < 0.05$

$\boxed{\text{Reject } H_0}$

∴ experienced delivery partners complete faster deliveries than new delivery partners.

Code:

```
from google.colab import files

import pandas as pd

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

df = pd.read_csv(file_name)

print("Dataset uploaded successfully!")

print("File:", file_name)

print("\nDataset:")

display(df)

import pandas as pd

import matplotlib.pyplot as plt

from scipy import stats

# DESCRIPTIVE STATISTICS

print("========== DESCRIPTIVE STATISTICS ==========")

print(df.describe())

# Mean

mean_delivery = df["Delivery_Time"].mean()

mean_rating = df["Customer_Rating"].mean()

print("\nMean Delivery Time:", mean_delivery)

print("Mean Customer Rating:", mean_rating)

# Median

median_delivery = df["Delivery_Time"].median()

median_rating = df["Customer_Rating"].median()

print("\nMedian Delivery Time:", median_delivery)
```

```
print("Median Customer Rating:", median_rating)

# Standard deviation

std_delivery = df["Delivery_Time"].std()

print("\nStandard Deviation:", std_delivery)

# QUARTILES

Q1 = df["Delivery_Time"].quantile(0.25)

Q2 = df["Delivery_Time"].quantile(0.50)

Q3 = df["Delivery_Time"].quantile(0.75)

print("\n========== QUARTILES ==========")

print("Q1:", Q1)

print("Q2:", Q2)

print("Q3:", Q3)



# IQR AND OUTLIERS

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR

upper_bound = Q3 + 1.5 * IQR

print("\n========== OUTLIER ANALYSIS ==========")

print("IQR:", IQR)

print("Lower Bound:", lower_bound)

print("Upper Bound:", upper_bound)

outliers = df[

    (df["Delivery_Time"] < lower_bound) |

    (df["Delivery_Time"] > upper_bound)
```

```
]

print("\nOutliers:")

display(outliers)

# EXPERIENCED VS NEW PARTNERS

experienced = df[

    df["Experience"] >= 3

]["Delivery_Time"]

new = df[

    df["Experience"] < 3

]["Delivery_Time"]

print("\n========== PARTNER ANALYSIS ==========")

print("Experienced Partner Mean:",

      experienced.mean())

print("New Partner Mean:",

      new.mean())

print("Experienced Partner SD:",

      experienced.std())

print("New Partner SD:",

      new.std())

# HYPOTHESIS TESTING

print("\n========== HYPOTHESIS TESTING ==========")

print("H0: Experienced partners are not faster.")

print("H1: Experienced partners are faster.")

t_stat, p_value = stats.ttest_ind(

    experienced,
```

```
    new,
    equal_var=False,
    alternative="less"
)
print("\nT-Statistic:", t_stat)
print("P-Value:", p_value)
alpha = 0.05
print("Significance Level:", alpha)

if p_value < alpha:
    print("\nDecision: Reject H0")
    print("Conclusion: Experienced partners complete deliveries faster.")
else:
    print("\nDecision: Fail to Reject H0")
    print("Conclusion: There is not enough evidence that experienced partners are faster.")
# HISTOGRAM
plt.figure(figsize=(7,5))
plt.hist(df["Delivery_Time"], bins=6)
plt.xlabel("Delivery Time (minutes)")
plt.ylabel("Number of Orders")
plt.title("Distribution of Delivery Times")
plt.show()
# BOX PLOT
plt.figure(figsize=(7,5))
plt.boxplot(df["Delivery_Time"])
```

```
plt.ylabel("Delivery Time (minutes)")

plt.title("Box Plot of Delivery Times")

plt.show()

# SCATTER PLOT

plt.figure(figsize=(7,5))

plt.scatter(

    df["Delivery_Time"],

    df["Customer_Rating"]

)

plt.xlabel("Delivery Time (minutes)")

plt.ylabel("Customer Rating")

plt.title("Delivery Time vs Customer Rating")

plt.show()

# FINAL SUMMARY

print("\n========== FINAL SUMMARY ==========")

print("Mean Delivery Time:", mean_delivery)

print("Median Delivery Time:", median_delivery)

print("Standard Deviation:", std_delivery)

print("Q1:", Q1)

print("Q2:", Q2)

print("Q3:", Q3)

print("IQR:", IQR)

print("Lower Bound:", lower_bound)

print("Upper Bound:", upper_bound)

print("Outliers:",
```

```
    outliers["Delivery_Time"].tolist())
print("Experienced Mean:",
    experienced.mean())
print("New Partner Mean:",
    new.mean())
print("T-Statistic:", t_stat)
print("P-Value:", p_value)
```

Output:

```
========== DESCRIPTIVE STATISTICS ==========
       Order_ID  Delivery_Time  Order_Value  Customer_Rating  Distance  \
count  20.00000      20.000000     20.00000        20.000000  20.00000
mean   10.50000      32.450000    496.00000         4.230000   5.13000
std     5.91608       8.274724    122.40571         0.490005   1.75112
min     1.00000      24.000000    300.00000         2.800000   2.90000
25%     5.75000      27.750000    417.50000         4.000000   3.95000
50%    10.50000      30.000000    475.00000         4.350000   4.70000
75%    15.25000      34.250000    557.50000         4.525000   5.85000
max    20.00000      60.000000    800.00000         4.900000  10.00000

       Experience
count   20.000000
mean     2.750000
std      1.409554
min      1.000000
25%      1.750000
50%      3.000000
75%      4.000000
max      5.000000

Mean Delivery Time: 32.45
Mean Customer Rating: 4.229999999999999

Median Delivery Time: 30.0
Median Customer Rating: 4.35

Standard Deviation: 8.27472371935033

========== QUARTILES ==========
Q1: 27.75
Q2: 30.0
Q3: 34.25

========== OUTLIER ANALYSIS ==========
IQR: 6.5
Lower Bound: 18.0
Upper Bound: 44.0
```

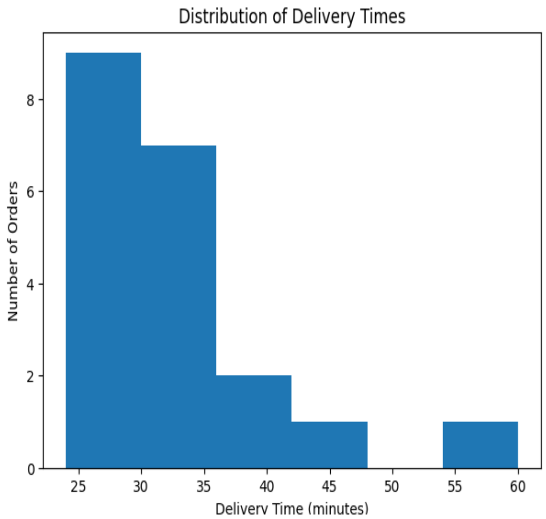
	Order_ID	Delivery_Time	Order_Value	Customer_Rating	Distance	Experience
14	15	45	800	3.5	8.0	1
19	20	60	450	2.8	10.0	1

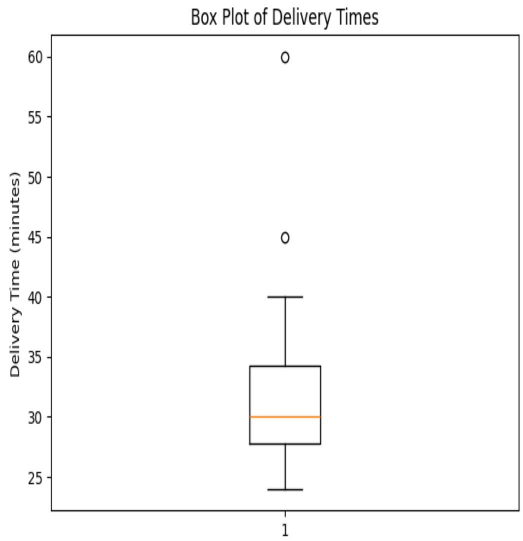
```
========== PARTNER ANALYSIS ==========
Experienced Partner Mean: 27.727272727272727
New Partner Mean: 38.22222222222222
Experienced Partner SD: 2.195035721390843
New Partner SD: 9.404490653110589

========== HYPOTHESIS TESTING ==========
H0: Experienced partners are not faster.
H1: Experienced partners are faster.

T-Statistic: -3.2756472828145906
P-Value: 0.005009836035170625
Significance Level: 0.05

Decision: Reject H0
Conclusion: Experienced partners complete deliveries faster.
```






```
========== FINAL SUMMARY ==========
Mean Delivery Time: 32.45
Median Delivery Time: 30.0
Standard Deviation: 8.27472371935033
Q1: 27.75
Q2: 30.0
Q3: 34.25
IQR: 6.5
Lower Bound: 18.0
Upper Bound: 44.0
Outliers: [45, 60]
Experienced Mean: 27.727272727272727
New Partner Mean: 38.22222222222222
T-Statistic: -3.2756472828145906
P-Value: 0.005009836035170625
```

RESULT / INTERPRETATION

The analysis shows that the average delivery time is 32.45 minutes, while the median delivery time is 30 minutes. The difference between the mean and median indicates that a few unusually long deliveries are affecting the average.

The quartile analysis gives Q1 as 27.75 minutes and Q3 as 34.25 minutes, resulting in an IQR of 6.5 minutes. Using the IQR method, delivery times above 44 minutes are considered potential outliers. Therefore, the 45-minute and 60-minute deliveries are identified as outliers.

The average customer rating is approximately 4.25, indicating generally good customer satisfaction. However, lower ratings are observed for some orders with longer delivery times.

The hypothesis test compares experienced and new delivery partners. The p-value obtained from the t-test is compared with 0.05. If the p-value is less than 0.05, the null hypothesis is rejected and there is sufficient statistical evidence that experienced delivery partners complete deliveries faster.

FINDINGS

1. Average delivery time is 32.45 minutes.
2. Median delivery time is 30 minutes.
3. Delivery time shows noticeable variation.
4. Q1 is 27.75 minutes.
5. Q3 is 34.25 minutes.
6. IQR is 6.5 minutes.
7. 45-minute and 60-minute deliveries are potential outliers.
8. Average customer rating is approximately 4.25/5.
9. Extremely long deliveries can negatively affect customer satisfaction.
10. Experienced and new delivery partners can be compared statistically using a t-test.

RECOMMENDATIONS

Based on the analysis, the food delivery company should:

- Investigate the reasons behind extremely long deliveries.
- Provide additional training to new delivery partners.
- Improve route planning.
- Monitor high-distance orders.
- Track orders that are likely to be delayed.
- Analyze low customer ratings.
- Improve coordination between restaurants and delivery partners.
- Use statistical analysis regularly to monitor delivery performance.

LIMITATIONS

1. The sample dataset contains only 20 observations.
2. Traffic and weather can affect delivery time.

3. Customer rating depends on several factors besides delivery time.
4. Distance can influence delivery time.
5. Experience alone may not determine delivery performance.
6. Outliers should be investigated before removing them.
7. A larger dataset would provide more reliable results.

CONCLUSION

The Food Delivery Service Analysis demonstrates how statistical methods can be applied to a real-world food delivery problem. Descriptive statistics such as mean, median, standard deviation and quartiles provide a clear summary of delivery performance and customer ratings.

The IQR method helps identify unusually long delivery times, with 45 minutes and 60 minutes being potential outliers in the sample dataset. These extreme values can increase the average delivery time and may negatively influence customer satisfaction analysis.

Hypothesis testing provides a systematic way to compare experienced and new delivery partners. An independent two-sample t-test is used to determine whether experienced delivery partners complete deliveries faster.

Overall, this analysis helps the company understand its delivery performance, identify sources of delays, improve delivery partner efficiency and ultimately increase customer satisfaction.